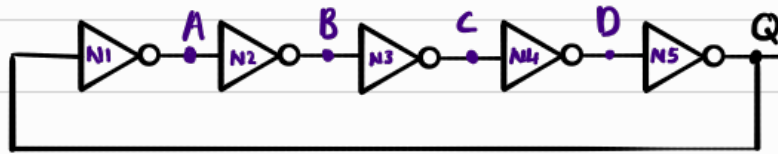
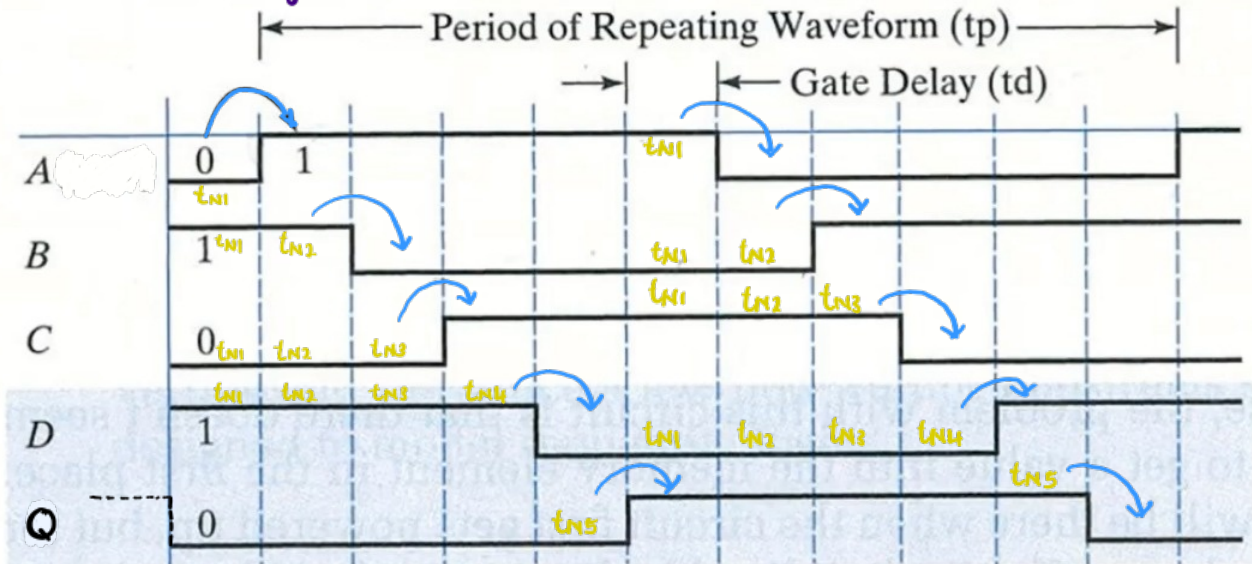


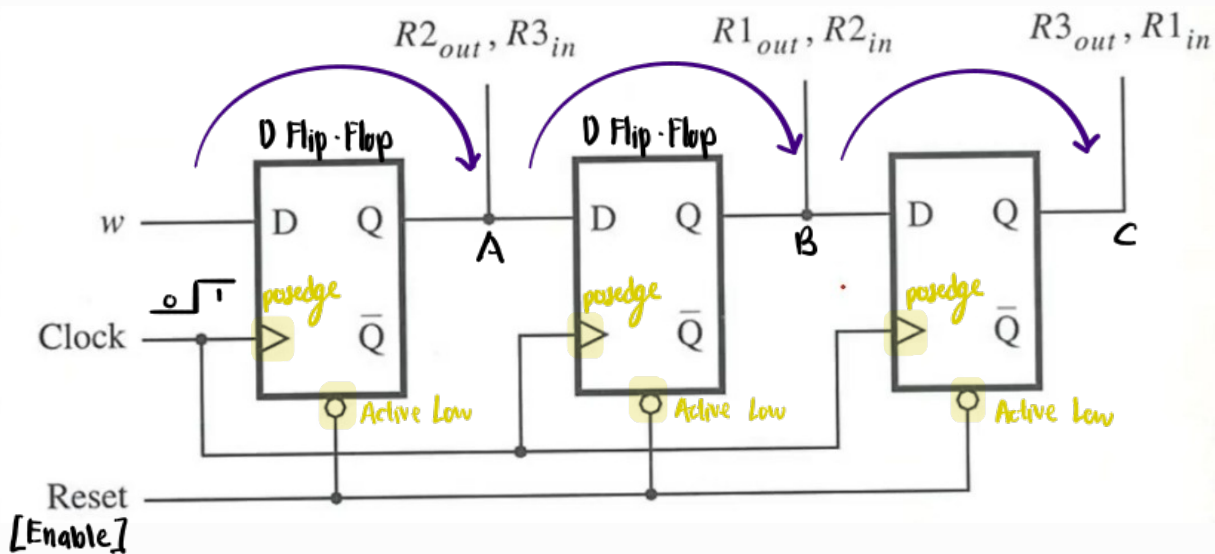
Ring Oscillator



Assume Q changes from 1 to 0 [$A=0, B=1, C=0, D=1$]

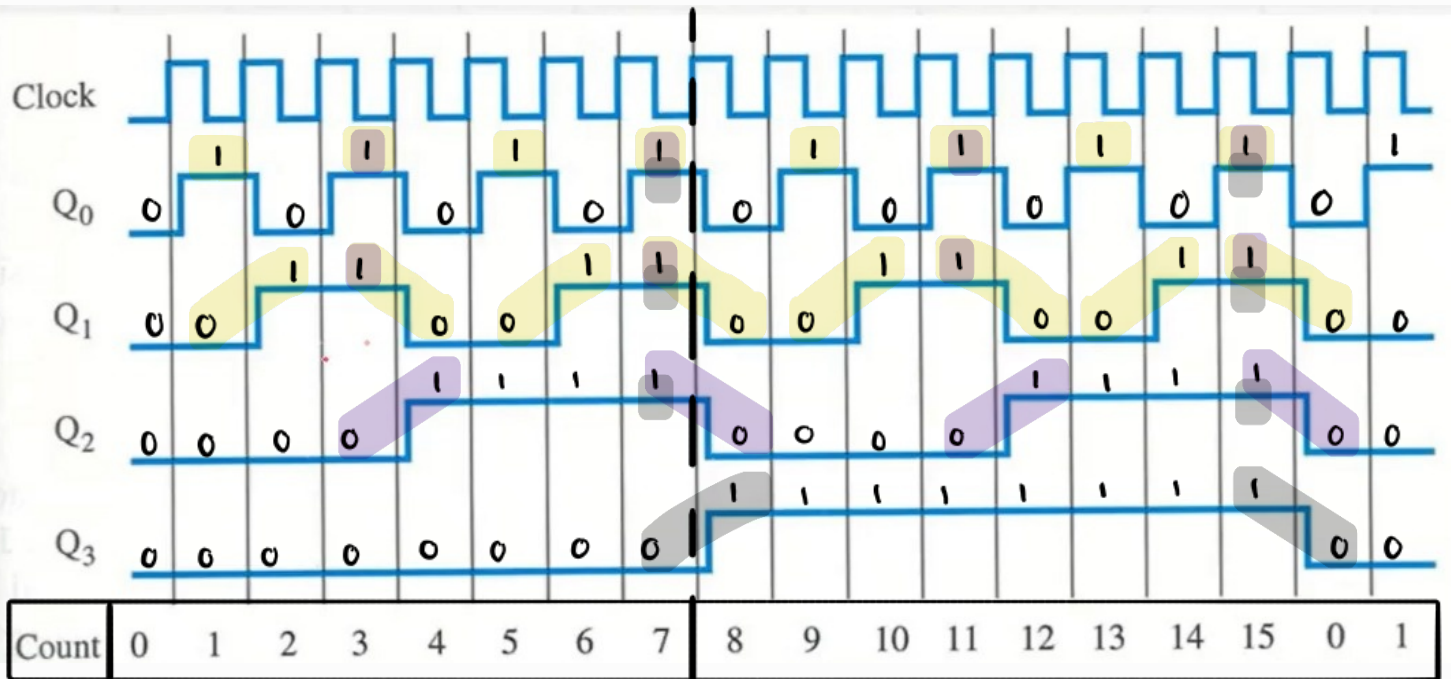
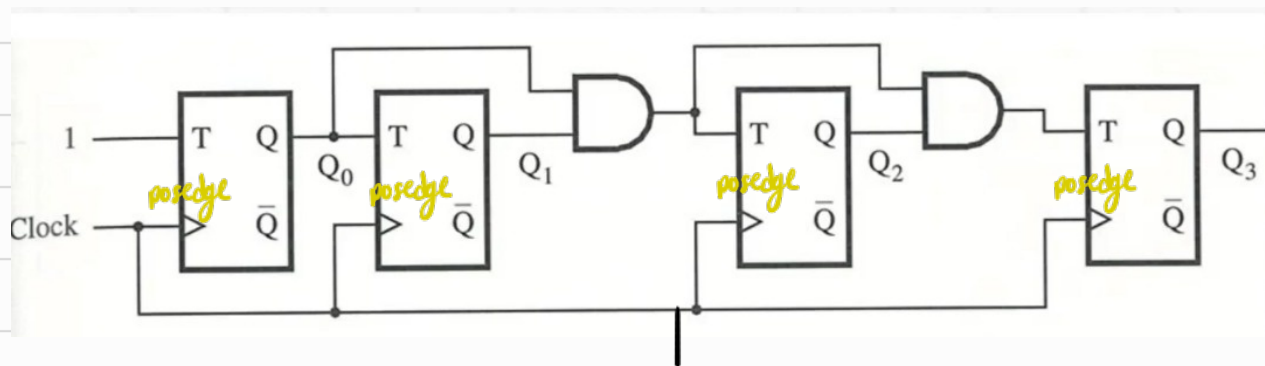


3 bit shift register



Every positive edge, each D flip-flop transmit $D \rightarrow Q$.
Causing all data in A, B, C to move to the right
by 1 step

4 bit counter using T flip-flops



One-hot sequence generator

Assuming initially $IN = 1$

